The Economic Cost of Depression: triangulation using panel and Mendelian Randomization analyses*

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Abstract

Depression, poor educational attainment, poor health, and unemployment form a correlated web that reduces human flourishing, but the causal links between them are unclear. We estimate the causal effect of depressive symptoms on educational and labour market outcomes by triangulating four research designs across economic, clinical, and genomic data. In UK longitudinal data (305,686 person-wave observations), a one standard-deviation (s.d.) rise in depressive symptoms lowers income in the following wave by 0.036 s.d. and employment by 0.044 s.d.; instrumenting symptoms with bereavement gives larger but less precise short-run estimates. Two genetic designs capture lifetime liability: two-sample Mendelian randomization on summary statistics from up to 2.1 million individuals, and a polygenic index instrument in the genotyped UKHLS subsample. In these designs, genetic liability to depression lowers income by 0.24–0.49 s.d., educational attainment by 0.14–0.46 s.d., and occupational status by 0.36–0.65 s.d. The education and occupation penalties fall almost entirely on women, while most of the income penalty for both sexes runs through employment rather than pay. The economic cost of depression accumulates over the life course, so treatment before labour-market entry may yield the largest economic returns.

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Depression affects an estimated 280 million people and is a leading cause of disability worldwide (World Health Organization, 2023; Moussavi et al., 2007). Together with anxiety, it costs the global economy around US$1 trillion a year in lost productivity (Chisholm et al., 2016). In the United Kingdom, the number of people aged 16–24 not in education, employment or training has passed one million for the first time in over a decade, and the share of this group reporting a work-limiting health condition has risen from 26% to 44% since 2015, mainly because of mental ill-health (Office for National Statistics, 2026; The Health Foundation, 2026). Governments now justify mental health spending partly in economic terms. A recent £343 million expansion of NHS community mental health centres, for example, was presented as a way to keep young people in education and adults in work (Department of Health and Social Care, 2026). These policies stake public money on a causal claim: that depression drives economic disadvantage, and not simply the reverse. Whether the claim holds, and whether the damage runs through education, occupation or employment, determines when in the life cycle treatment offers the greatest economic return. This paper tests the claim and traces the channels.

There are good theoretical reasons to expect such an effect. Depression has been conceptualised as "pessimism about the returns to effort" (De Quidt and Haushofer, 2016), which can lead to partial or complete withdrawal from the challenges of education and the labour market. The empirical difficulty is that the correlations motivating the claim cut both ways: people with lower socioeconomic status are significantly more likely to experience depression (Ettner et al., 1997), and the causal effect of income on depression is well documented in experimental (Ridley et al., 2020) and genetically informed studies (Marchi et al., 2024; Demange et al., 2024). The reverse effect, of depressive symptoms on income, education, and occupational attainment, is harder to establish: depressive symptoms cannot be experimentally assigned, and observational designs cannot easily separate their effects from those of childhood environment, cognitive ability, and other correlated factors.

To address these limitations, we combine economic, social, clinical and genomic data sources to triangulate an estimate of the causal effect of depressive symptoms on four outcomes of interest: income, educational attainment, occupational attainment, and employment. We used two-way fixed-effects (TWFE) panel models on 305,686 person-wave observations in the British Household Panel Survey (BHPS) and its successor, the UK Household Longitudinal Study (UKHLS), exploiting within-individual variation in depressive symptoms over time to predict later labour market outcomes. We used phenome-wide association clustering (Darrous et al., 2024) and two-sample Mendelian randomization (Sanderson et al., 2022) on genome-wide association summary statistics from cohorts of 249,000 to 2.1 million individuals (Baselmans et al., 2019; Kweon et al., 2024; Okbay et al., 2022; Ko et al., 2022; Akimova et al., 2025), exploiting the random assignment of genetic alleles at conception to estimate causal effects. We used bereavement as an instrumental variable for depressive symptoms in 258,048 person-year observations in the same UK panel, treating the loss of a close relative or friend as a shock to depressive symptoms unrelated to prior labour market trajectories. And we used two-stage least squares on individual polygenic index values capturing genetic propensity towards depression in the genotyped UKHLS subsample, which estimates the effect of lifetime liability to depression on the same outcomes.

By triangulating findings across these four modelling strategies we obtain estimates of the effect of depressive symptoms on educational and labour market outcomes that are robust

to violations of the assumptions of any one method.

1 Results

1.1 Depressive symptoms reduce income, employment and occupational status

Figure 1 summarises the research design and Figure 2 reports the headline estimates. Each point is the estimated effect of a one standard-deviation (s.d.) increase in depressive symptoms on a standardised economic outcome. The two within-person designs (panel model and bereavement instrument) identify short-run effects of a change in symptoms among working-age adults; the two genetic designs (two-sample Mendelian randomization, MR, and the polygenic index, PGI, instrument) identify the effect of lifetime liability, which also operates through education and early occupational sorting. The two horizons are different quantities and are not expected to coincide.

All four designs find that depressive symptoms reduce income. In the panel model, a one-s.d. rise in lagged depressive symptoms lowers income in the following wave by 0.036 s.d. (95% CI $[-0.040, -0.033]$; $n = 305{,}686$ person-waves). Instrumenting symptoms with non-spousal bereavement gives a larger but less precise short-run estimate of -0.32 s.d. ($[-0.63, 0.00]$). The two lifetime estimates are an order of magnitude larger than the panel estimate: -0.49 s.d. ($[-0.72, -0.26]$) in two-sample MR and -0.24 s.d. ($[-0.27, -0.21]$) with the PGI instrument.

Three of the four designs also find that depressive symptoms reduce the probability of employment: by 0.044 s.d. ($[-0.048, -0.040]$) in the panel model, 0.50 s.d. ($[-0.86, -0.13]$) with the bereavement instrument and 0.42 s.d. ($[-0.48, -0.37]$) with the PGI instrument. The two-sample MR estimate for employment is the one headline result that does not follow this pattern: it is close to zero and imprecise (0.06 s.d., $[-0.25, 0.37]$), with a confidence interval that contains the panel and PGI point estimates. When the within-family MR is estimated on its genome-wide instrument of 80–91 variants rather than the six-variant brain-eQTL instrument (below), the employment effect is negative and significant (-0.08 s.d., $[-0.16, -0.01]$), so we read the six-variant null as a limit of power for a binary outcome rather than as evidence against an employment effect.

Occupational status and education separate the short-run from the lifetime designs. Neither within-person design detects any effect on occupational status (0.000 s.d., $[-0.003, 0.003]$ in the panel; 0.03 s.d., $[-0.26, 0.32]$ with the bereavement instrument), whereas both genetic designs find large negative effects (-0.65 s.d., $[-1.00, -0.30]$ in two-sample MR; -0.36 s.d., $[-0.43, -0.28]$ with the PGI). Genetic liability to depression also lowers educational attainment, by 0.46 s.d. ($[-0.75, -0.17]$) in two-sample MR and 0.14 s.d. ($[-0.19, -0.09]$) with the PGI; education is fixed over the working-age window and so cannot be studied in the within-person designs. Taken together, depression appears to shape occupational standing through the education and sorting decisions that precede working life, rather than through movement between occupations during it.

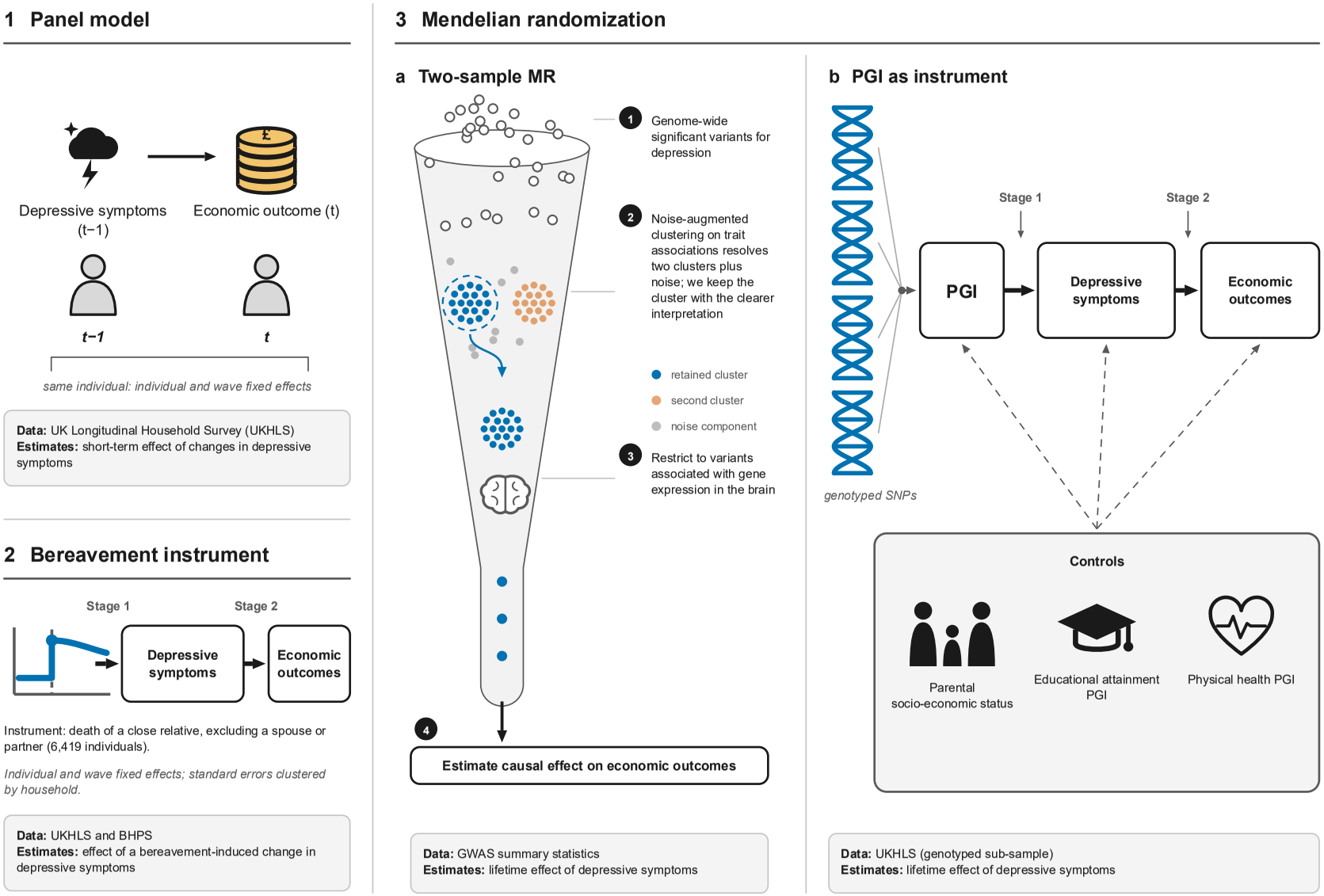


Figure 1: Research design for panel and Mendelian Randomization analyses

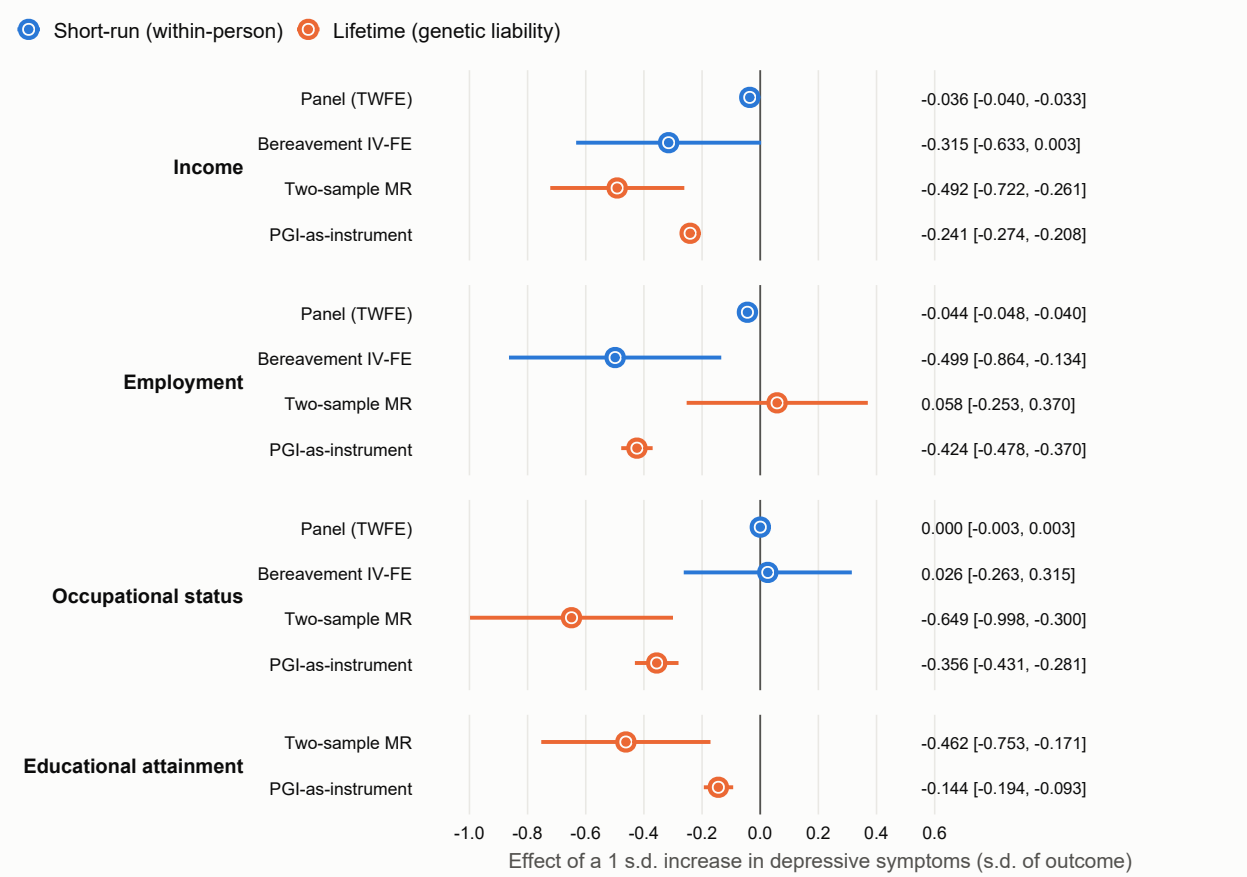


Figure 2: **Effect of depressive symptoms on economic outcomes across four identification strategies.** Points are the estimated effect of a one standard-deviation increase in depressive symptoms; bars are 95% confidence intervals. Blue marks the two within-person designs, which identify short-run effects among working-age adults; orange marks the two genetic designs, which identify effects of lifetime liability and so also capture pathways through education and early occupational sorting. The two horizons are distinct estimands and are not expected to coincide. The bereavement IV-FE design instruments depressive symptoms with the death of a *non-spouse* relative, friend or colleague; spouse deaths are excluded because they directly change household labour supply. Occupational status is the ISEI (Ganzeboom) socio-economic index throughout, mapped from ISCO-88 in the individual-level designs and taken from the Akimova et al. (2025) ISEI GWAS for two-sample MR. Educational attainment is fixed within person over the working-age window and is therefore not estimable in the panel designs.

1.2 Lifetime effects on education and occupation are concentrated among women

Figure 3 reports sex-specific estimates. For income and employment the sexes are similar. The PGI income effects are -0.25 s.d. ($[-0.29, -0.20]$) for women and -0.23 s.d. ($[-0.28, -0.19]$) for men (difference $p = 0.73$), and the PGI employment effects are -0.45 s.d. and -0.38 s.d. respectively ($p = 0.17$). In the panel model the short-run income and employment effects are slightly larger for men (-0.023 and -0.046 s.d.) than for women (-0.018 and -0.032 s.d.). With the bereavement instrument the income and employment effects are significant for women (-0.50 s.d., $[-0.92, -0.08]$; -0.66 s.d., $[-1.16, -0.16]$) but not for men, although the male estimates are imprecise and the confidence intervals overlap.

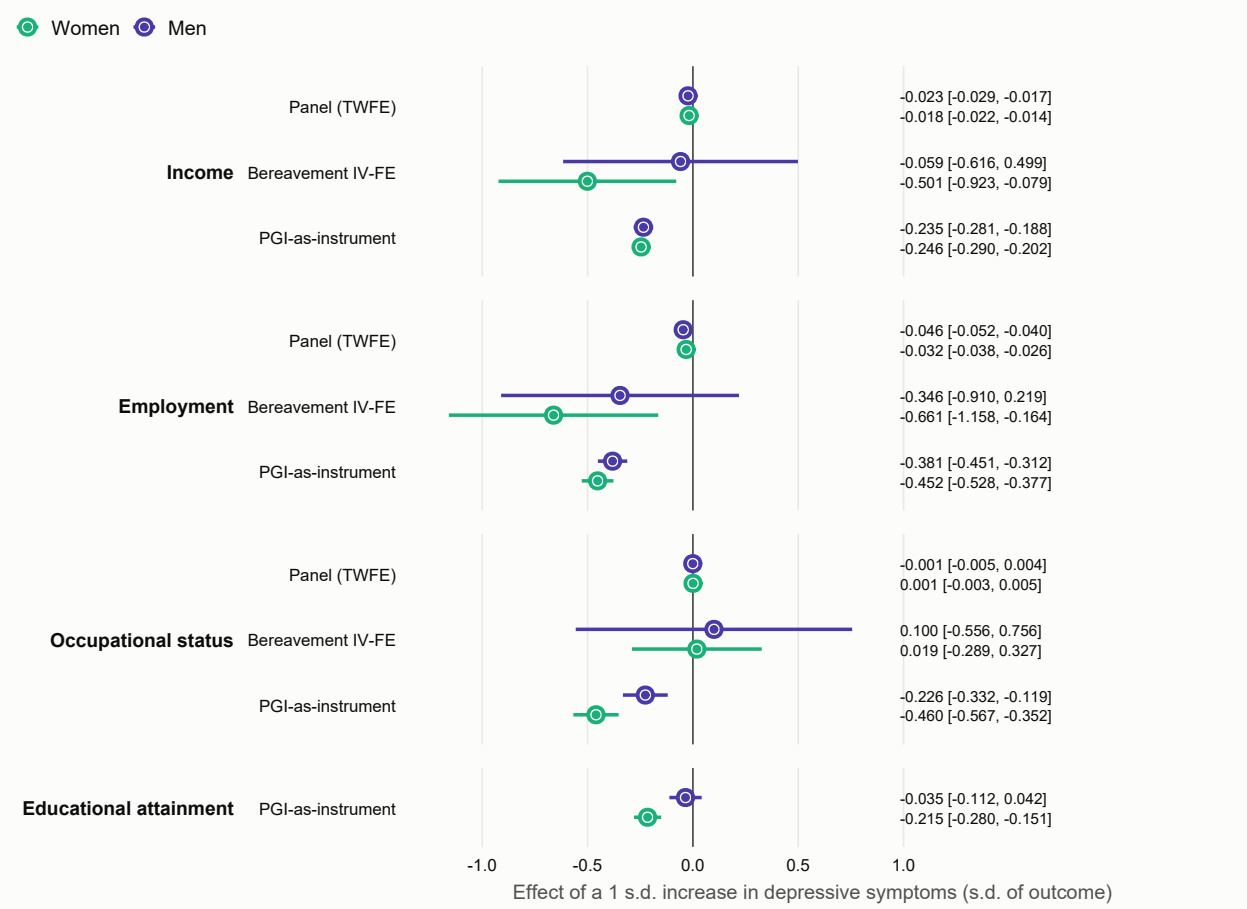


Figure 3: **Sex-specific effects of depressive symptoms, by outcome and design.** Points are estimates; bars are 95% confidence intervals. The bereavement IV-FE design instruments depressive symptoms with the death of a *non-spouse* relative, friend or colleague. Occupational status is the ISEI (Ganzeboom) index in all three designs. Two-sample MR is omitted: sex-stratified outcome GWAS exist only for income, and at the six-SNP instrument that estimate cannot separate the sexes. Educational attainment is shown for the PGI design alone, being fixed within person over the working-age window. Sex differences are significant for educational attainment ($p < 0.001$) and occupational status ($p = 0.002$), but not for income ($p = 0.73$) or employment ($p = 0.17$).

The sexes diverge on the two outcomes determined early in the life course. Genetic liability to depression lowers women's educational attainment by 0.22 s.d. ($[-0.28, -0.15]$) but has no detectable effect on men's (-0.04 s.d., $[-0.11, 0.04]$; difference $p < 0.001$), and it lowers women's occupational status by twice as much as men's (-0.46 s.d., $[-0.57, -0.35]$ against -0.23 s.d., $[-0.33, -0.12]$; $p = 0.002$). These sex-specific results are robust to measuring depressive symptoms with the GHQ-12 Likert scale or the SF-12 mental component score (Supplementary Section S5).

1.3 Most of the income effect runs through employment

Figure 4 decomposes the income effect by adding potential mediators in sequence and recording the share of the total effect each one absorbs. In the PGI design, labour-market attachment dominates: employment history (lifetime employment rate, long-term sickness and job changes) and current employment status together account for 83% of the effect among women (55% and 28%) and 75% among men (44% and 31%). Education explains 10% of the effect among women and almost none among men; first occupation adds little for either sex. Working hours absorb a further 10% of the effect among men. Once all mediators are included, the remaining direct effect is small: about 4% of the total among women and 14% among men.

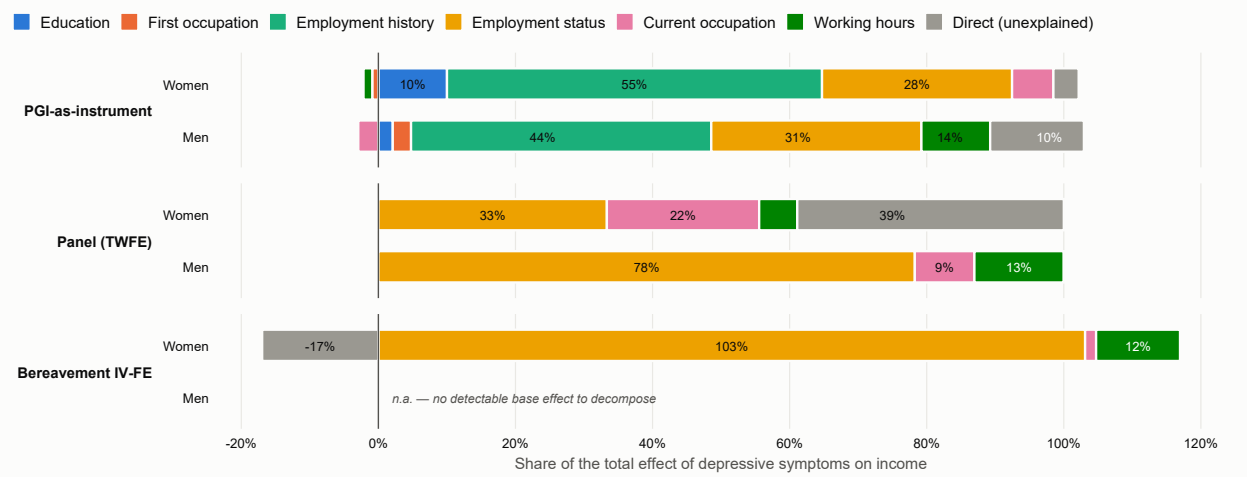


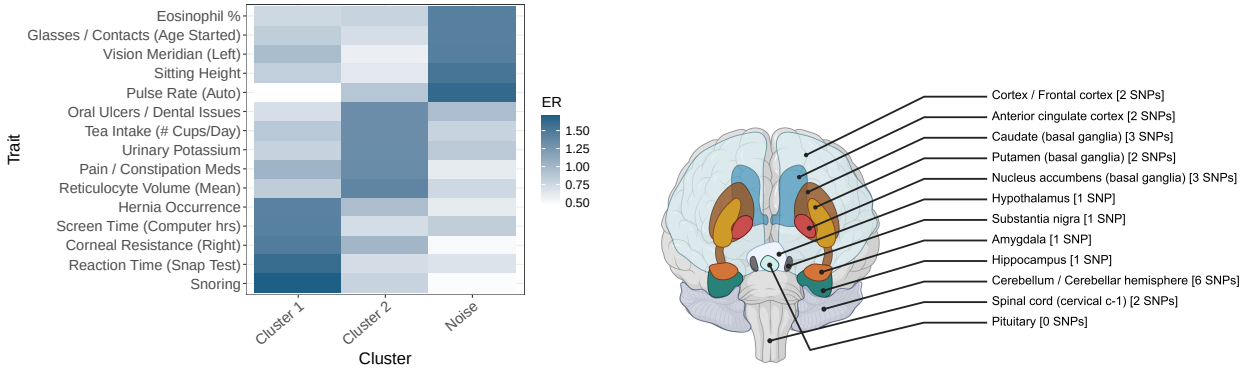
Figure 4: **Mediation of the income effect, by sex.** Each bar splits the total effect of depressive symptoms on income into the share absorbed as each mediator is added in sequence, plus the direct effect no observed mediator explains; shares are $(\beta_{k-1} - \beta_k)/\beta_0$ with the final segment β_K/β_0. The bereavement IV-FE design instruments depressive symptoms with the death of a *non-spouse* relative, friend or colleague. Education, first occupation and employment history are absorbed by the individual fixed effect in the two within-person designs, so they appear only for the PGI design. Segments left of zero denote suppression rather than an explained share. For the bereavement IV-FE design the mediators over-explain the effect among women, because conditioning on employment conditions on a variable the instrument itself moves; men are not decomposed because the base effect is indistinguishable from zero.

The short-run income effect in the panel model is also mostly an employment effect, but

the pattern differs by sex. Among men, employment status accounts for 78% of the effect, changes in occupation for 9% and working hours for 13%, leaving nothing unexplained. Among women, employment status accounts for 33% and occupation for 22%, with 39% of the effect not explained by employment, occupation or hours; this residual is consistent with within-job changes in pay or productivity that the mediators do not capture. Among bereaved women the employment channel accounts for the whole of the income effect; the male bereavement estimate is too imprecise to decompose.

1.4 Robustness of the genetic estimates

The two-sample MR instrument is the subset of depression-associated variants that share a coherent biological profile. Figure 5a shows the five traits most enriched in each NAVMIX cluster. The sedentary cluster used throughout the main text loads on snoring, reaction time, corneal resistance, screen time and hernia, a neurological and somatic profile consistent with variants acting on depression through the brain. The second cluster loads on a heterogeneous mix of blood-cell, dietary and medication traits (reticulocyte volume, paracetamol use, tea intake) that are more plausibly confounded by socioeconomic status or health behaviours, and the noise cluster on traits with no coherent link to depression (pulse rate, sitting height, vision). Figure 5b shows where the depression-associated variants in the sedentary cluster regulate gene expression in GTEx brain tissue. Expression is concentrated in the cerebellum (all six variants), with two to three variants each in the frontal and cingulate cortex and in the basal ganglia (caudate, putamen and nucleus accumbens), regions repeatedly implicated in depression by neuroimaging and post-mortem studies.



(a) Enrichment ratio for the top five traits in each cluster.

(b) Brain regions in which depression-associated variants regulate gene expression (GTEx).

Figure 5: **Cluster-specific enrichment and brain gene expression of depression-associated variants.** (a) Enrichment ratio (average squared standardised effect in the cluster relative to the average across clusters; Methods) for the five most enriched traits in each NAVMIX cluster. (b) Number of depression-associated variants that are expression quantitative trait loci (eQTLs) in each GTEx brain region; a variant can be an eQTL in several regions.

Figure 6 reports sensitivity analyses for the two-sample MR estimates. For income and occupational status, the weighted-median, weighted-mode and MR-PRESSO estimators give

effects of the same sign and similar size as the inverse-variance-weighted (IVW) estimate (-0.49 to -0.59 s.d. for income; -0.53 to -0.65 s.d. for occupational status). For education the pleiotropy-robust estimators are about half the IVW estimate and their confidence intervals include zero (-0.23 s.d., $[-0.55, 0.09]$ for the weighted median), so the education estimate should be read as less secure than the income and occupation estimates. Steiger tests support the assumed direction of causation for income, education and employment ($p \leq 2.8 \times 10^{-4}$); for occupational status the test is directionally correct but not significant ($p = 0.13$), so the ISEI estimate rests on the assumption that liability precedes occupational sorting rather than on a statistical test of it.

Two checks address the threats specific to a two-sample design built on UK Biobank GWAS. First, correcting for sample overlap, winner's curse and weak instruments with MR-LAP (Mounier and Kutalik, 2023) preserves the sign and inference of every estimate; the corrected effects are larger in magnitude (income -0.81 s.d., $[-1.25, -0.37]$; education -0.80, $[-1.35, -0.26]$; occupational status -1.19, $[-1.89, -0.48]$). Because sample overlap biases IVW *towards* the observational association, a correction that moves estimates away from zero reflects MR-LAP's winner's-curse term, which over-corrects a small, hand-selected instrument; when MR-LAP selects its own genome-wide instrument the correction is modest (Supplementary Section S9). Second, to rule out effects that run through the family a respondent grew up in rather than through their own depression, we re-estimate the MR using the within-sibship GWAS of Guan et al. (2025). Direct (within-family) and population SNP–exposure effects give essentially the same answer for all four outcomes (income -0.25 s.d. against -0.23; education -0.32 against -0.34; occupational status -0.27 against -0.31; employment -0.08 against -0.10), and equality of the two arms is not rejected for any outcome ($p \geq 0.21$; Supplementary Section S13).

The placebo instrument, built from variants whose multi-trait profile is incoherent, has no significant effect on any outcome. The headline instrument behaves as it should against controls: it raises sibling depression, the positive control most directly capturing depression liability ($p = 4 \times 10^{-6}$), and lowers cognitive ability ($p = 3 \times 10^{-5}$), while the self-harm and friend-dissatisfaction coefficients are in the expected direction but not individually significant ($p = 0.17$ and $p = 0.16$); no negative control — hair colour, skin colour or birth location — is significant after Bonferroni correction (Supplementary Figures S6–S7). The income estimate is also unchanged when UK Biobank is excluded from the outcome GWAS (Supplementary Figure S9). Results for the broader 14-SNP brain-expressed instrument and for each NAVMIX cluster are reported in Supplementary Section S8.

The PGI instrument is strong (first-stage $F >$ 1,000), the Anderson–Rubin test rejects a zero reduced-form effect ($p < 0.001$), the estimate survives the tF correction of Lee et al. (2022), and a leave-one-household-out jackknife shows that no single household drives the result (Supplementary Table S38). PGI estimates condition on parental education and occupation, ten genetic principal components, and polygenic indices for educational attainment and physical health, so that the instrument's effect is not carried by correlated genetic endowments or family background.

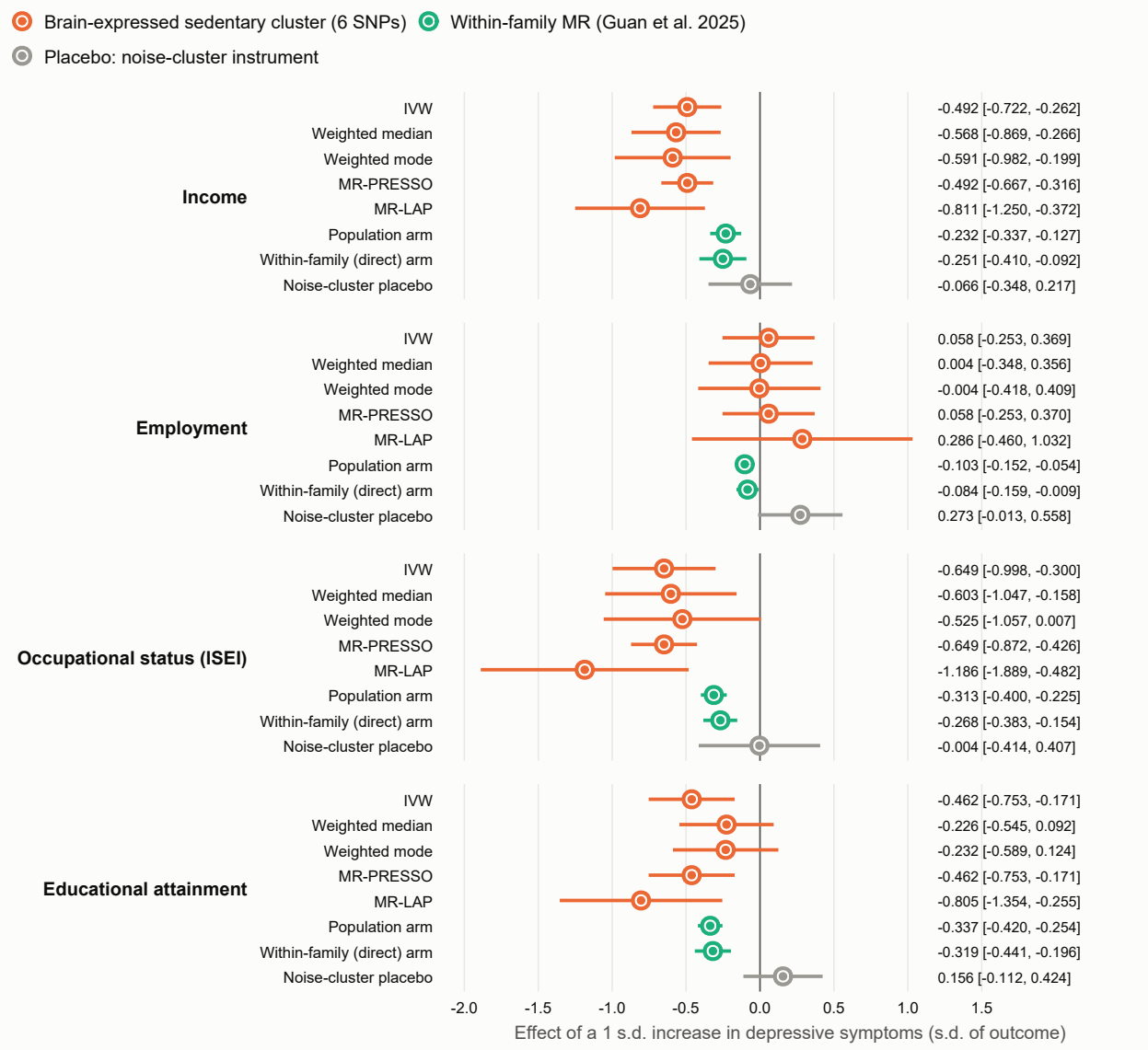


Figure 6: **Two-sample MR robustness for the four primary outcomes.** Effect of a one-s.d. increase in genetically proxied depressive symptoms; bars are 95% CIs. Orange: the 6-SNP brain-expressed sedentary-cluster instrument under IVW, weighted median, weighted mode, MR-PRESSO and the MR-LAP sample-overlap correction (Mounier and Kutalik, 2023). Green: within-family MR using the family-based GWAS of Guan et al. (2025), with population and direct (within-family) SNP–exposure effects (debiased IVW; income is individual income). Grey: placebo noise-cluster instrument (weighted median). Numerical values and diagnostics are in Supplementary Table S20.

1.5 Robustness of the within-person estimates

The panel estimates are unchanged when the SF-12 mental component score replaces the GHQ-12 as the measure of depressive symptoms (Supplementary Section S3). Because depression is a reversible rather than absorbing state, staggered-adoption diagnostics do not apply directly; a distributed lead–lag specification with three leads and three lags of the exposure shows no pre-trend (all lead coefficients are near zero) and a response that decays over three waves, so the single-lag benchmark is a conservative reading of the short-run effect (Supplementary Section S4). For the bereavement instrument (n = 258,048; 6,040 bereaved; first-stage $F = 21.4$), event-study estimates show no pre-bereavement trend in either depressive symptoms or economic outcomes (Supplementary Figure S2). Deaths of a spouse are excluded because they change household labour supply directly, and an over-identification test rejects the serious illness of a close relative as an additional instrument ($p = 0.027$), consistent with a direct caregiving channel (Supplementary Section S6).

2 Discussion

Across four identification strategies with largely uncorrelated biases, we find that depressive symptoms causally and robustly reduce income and educational attainment, with mixed negative effects on occupational status and employment. These four outcomes describe the one million UK young people currently not in education, employment or training (NEET). A previous meta-analysis by Gariépy et al. (2022) finds NEET young people at elevated risk of mood and anxiety disorders, substance use and suicidality, with a suggestive causal link between depression and later NEET status. Our findings support a causal role for depressive symptoms in creating and sustaining NEET status, and demonstrate a consistent causal effect across multiple cohorts and study designs combining panel, clinical, and genomic data. The income finding sits in tension with, without directly contradicting, Mendelian randomization research on depression and poverty, which finds that major depressive disorder does not significantly increase the risk of poverty (Marchi et al., 2024). We note that poverty and income are related but not identical measures, and the effects of depressive symptoms may be non-linear as income approaches the poverty line; future work could test this hypothesis in an appropriately powered biobank.

We find that long-term exposure to depression risk confers a substantially larger reduction in educational attainment and occupational status than short-term exposure. The same pattern is documented for other exposures with cumulative effects, such as lifetime exposure to LDL cholesterol (Ference et al., 2012), and indicates a detrimental effect of depression that compounds throughout the life course. Our mediation analysis supports a cumulative effect: accumulated employment history plays a larger role even than current employment in explaining the effect of depression on income. Together, these two findings point to the relative importance of mental health interventions early in the life course, as later-life shocks cannot affect educational attainment and appear not to affect occupational status.

We find suggestive evidence that the effects of depression on labour market outcomes differ by sex. Genetic liability to depression costs women and men the same income (-0.25 against -0.23 s.d.; $p = 0.73$) and the same employment probability ($p = 0.17$), but only

among women does it measurably lower educational attainment (-0.22 against -0.04 s.d.; $p < 0.001$), and it lowers women's occupational status by twice as much as men's (-0.46 against -0.23 s.d.; $p = 0.002$). Lifetime liability appears to disproportionately reduce women's human-capital formation and occupational sorting, whereas among men a similar income loss is delivered almost entirely through weaker attachment to work at a given level of schooling and occupation.

Several limitations qualify these conclusions. All discovery GWAS and individual-level analyses use participants of European-like ancestry, so the estimates should not be assumed to transport to other ancestral groups; replication in diverse cohorts is a priority. The genetic and within-person estimands differ in exposure timing, so the gap between them is informative about timing rather than evidence that either is biased. Because UKHLS and the discovery GWAS are predominantly British, the magnitude of the income penalty may differ where sick-pay, welfare and disability-benefit institutions differ; the qualitative finding should generalise, but magnitudes should be re-estimated elsewhere. Finally, the two-sample MR null for employment has a confidence interval containing both the panel and PGI estimates; with a binary outcome and a six-variant instrument this reflects limited power rather than evidence against an employment effect, as the negative within-family estimate on 80–91 variants indicates.

Based on the size of the lifetime estimates, and the concentration of sex differences in education and occupational sorting, we cautiously suggest that mental-health intervention earlier in the life cycle could have larger cumulative economic effects than intervention in working age. Further research could test this hypothesis experimentally or through meta-analysis of existing age-linked interventions.

The two estimands do, however, bound the cost of a concrete change. Between 2015–19 and 2022–24, mean depressive symptoms in UKHLS rose by about 0.2 s.d. among adults under 45 and by half that above 45 (Supplementary Section S15). If that rise proves transitory, the panel coefficient implies roughly 100,000 fewer people in work across the working-age population and £3 billion a year in lost earnings at 2024 prices; if it proves permanent, the PGI coefficient implies close to one million fewer in work and £30 billion a year, with three-quarters of either figure falling on adults under 45. These are partial-equilibrium figures: they hold wages fixed, whereas a withdrawal of labour on this scale would raise wages and draw some workers back in, so that the general-equilibrium employment loss would be smaller than the sum of individual responses (Heckman et al., 1998). The range is wide because the persistence of the deterioration is the quantity that matters, and it is not yet known; but even the conservative reading is of the same order as recent mental-health spending commitments, and the concentration among the young follows directly from where symptoms have risen. Large within-family genotyped cohorts, and designs estimating both directions of the depression–income relationship jointly, are the natural next steps.

3 Methods

3.1 Study design and identifying assumptions

We estimate the effect of depressive symptoms on four economic outcomes — income, employment, occupational status and educational attainment — with four designs whose biases are largely independent (Figure 1). Two within-person designs use the British Household Panel Survey (BHPS) and UK Household Longitudinal Study (UKHLS): a two-way fixed-effects (TWFE) panel model, and an instrumental-variables fixed-effects (IV-FE) design that uses the timing of bereavement as an instrument for depressive symptoms. Two genetic designs use the random allocation of alleles at conception: two-sample Mendelian randomization (2SMR) on genome-wide association study (GWAS) summary statistics, and two-stage least squares (2SLS) with a depression polygenic index (PGI) as the instrument in the genotyped UKHLS subsample. The within-person designs identify the short-run effect of a change in symptoms among working-age adults; the genetic designs identify the effect of lifetime liability, which also operates through education and early occupational sorting.

All instrumental-variable estimates, genetic or otherwise, rest on three assumptions: *relevance* (the instrument is associated with depressive symptoms), *independence* (the instrument is not associated with confounders of the symptom–outcome relationship) and *exclusion* (the instrument affects the outcome only through depressive symptoms). For genetic instruments, independence is threatened by population stratification, assortative mating and dynastic effects, and exclusion by horizontal pleiotropy; for bereavement, independence requires that its timing is as good as random conditional on individual and year fixed effects, and exclusion that it affects economic outcomes only through mental health. A fourth assumption, monotonicity, is required to interpret the estimates as local average effects. The specific checks we use for each design are described below, and the Discussion sets out what each design can and cannot rule out.

3.2 Data

3.2.1 BHPS and UKHLS

The BHPS (1991–2008, about 10,000 individuals) and its successor UKHLS, also known as Understanding Society (2009–2023, about 40,000 households), are annual household panels that follow the same individuals over time (University of Essex, Institute for Social and Economic Research, 2025). We pool the two into a single panel of person-waves. Depressive symptoms are measured with the 12-item General Health Questionnaire (GHQ-12), a screening instrument for psychological distress asked of every adult in every wave (Anjara et al., 2020); we use the Likert score (0–36), rank-normalised to mean 0 and s.d. 1 and signed so that higher values indicate worse mental health, and the 0–12 caseness score for the IV-FE design. The 12-item Short Form Survey mental component score (SF-12 MCS), available only in UKHLS (Ware et al., 1996; Salyers et al., 2000; Gill et al., 2007), is used as an alternative exposure (correlation with the GHQ-12 of -0.77). Item wording for both instruments is given in Supplementary Section S1.

Income is the log of real gross monthly labour income (2015 prices, CPIH-deflated), standardised; employment is an indicator for paid employment or self-employment, standardised;

occupational status is the International Socio-Economic Index (ISEI) of Ganzeboom et al. (1992), assigned from the ISCO-88 code of the current job and standardised (Supplementary Section S12 reports alternative scales); educational attainment is the highest qualification held. Approximately 10,000 UKHLS participants were genotyped on the Illumina HumanCoreExome array and imputed to the UK10K reference panel, which allows the PGI design. Sample flows, descriptive statistics for each analysis sample and a comparison of the genotyped subsample with the full panel are reported in Supplementary Section S2.

3.2.2 GWAS summary statistics

All GWAS are of European-ancestry samples; sources, sample sizes, covariate adjustments and accession identifiers are listed in Supplementary Table S18. The exposure GWAS is the N-weighted multivariate meta-analysis of depressive symptoms of Baselmans et al. (2019) (N = 2,083,151, excluding 23andMe). Outcome GWAS are: the income factor of Kweon et al. (2024) (individual, household and occupational income; N = 668,288; versions excluding UK Biobank are used in sensitivity analyses), educational attainment from Okbay et al. (2022) (EA4, N = 765,283), occupational status (ISEI) from Akimova et al. (2025) (UK Biobank, $N \approx$ 272,000), and employment status from UK Biobank (OpenGWAS ukb-b-6740, N = 461,242). Positive-control outcomes are self-harm, sibling depression, friend satisfaction, Townsend deprivation and cognitive ability; negative controls are hair colour, skin colour, ease of tanning and the north and east coordinates of current residence and birthplace. All source GWAS adjusted for age, sex and genetic principal components; the exposure and outcome discovery samples are similar in age structure and ascertainment, so SNP–exposure associations are expected to transport to the outcome samples. UK Biobank contributes to the exposure GWAS and to most outcome GWAS; we quantify and correct for the resulting sample overlap as described below.

3.3 Panel model (TWFE)

We estimate

$$y_{i,t} = \beta\, dep_{i,t-1} + \delta \mathbf{X}_{i,t} + b_i + d_t + \epsilon_{i,t}, \tag{1}$$

where $y_{i,t}$ is the standardised outcome, $dep_{i,t-1}$ is lagged standardised depressive symptoms, $\mathbf{X}_{i,t}$ is a quadratic in age, and b_i and d_t are individual and wave fixed effects (Wooldridge, 2010). The one-wave lag addresses reverse causality from contemporaneous earnings shocks to mental health; the contemporaneous specification, and a richer control set adding marital status and number of children interacted with sex, are reported in Supplementary Section S3. Standard errors are clustered by household. The sample is working-age adults (25–64) in active labour-force states (employed, self-employed, unemployed, family care or long-term sick), giving n = 305,686 person-waves for income and employment and 233,558 for occupational status (employed person-waves with a valid ISCO-88 code). Educational attainment is fixed within person over this window and is not a panel outcome. Because depression is a reversible state that individuals enter and leave repeatedly, the staggered-adoption concerns raised for TWFE with absorbing binary treatments (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2020) do not apply directly; we assess dynamics and pre-trends with a distributed lead–lag specification that includes three leads

and three lags of the exposure (Supplementary Section S4). Mediation is assessed by adding employment status, occupation and working hours sequentially to Eq. 1 and recording the attenuation of β.

3.4 Bereavement as an instrument (IV-FE)

The second within-person design instruments standardised GHQ-12 caseness with an indicator that switches on from the first wave in which the respondent reports the death of a close relative other than a spouse, or of a close friend, in an instrumented difference-in-differences framework (de Chaisemartin and D'Haultfœuille, 2018; Frijters et al., 2014; Hudson et al., 2017). Models include individual and year fixed effects and the controls in Eq. 1; standard errors are clustered by household. The sample is person-years at ages 25–55 between 1991 and 2019 in the same labour-force states as the panel model (n = 258,048; 6,040 bereaved); the first-stage F-statistic is 21.4 for income and employment and 24.0 for occupational status. Spouse deaths are excluded because they change household labour supply directly. Serious illness of a close relative was considered as a second instrument and rejected by a Sargan over-identification test ($p = 0.027$), consistent with a caregiving channel that violates exclusion. We assess parallel trends with Sun–Abraham event-study estimates of the first stage and the reduced form (Sun and Abraham, 2021), and report the Wald-DID-IV estimator built from them alongside the TWFE-IV headline (Supplementary Section S6).

3.5 Two-sample Mendelian randomization

3.5.1 Instrument selection

Candidate variants are those reaching genome-wide significance ($p < 5 \times 10^{-8}$) in the exposure GWAS, clumped to independence ($r^2 < 0.001$ within 10,000 kb) against the European subsample of the 1000 Genomes reference panel (The 1000 Genomes Project Consortium, 2015; Purcell et al., 2007). Palindromic SNPs are aligned by allele frequency or excluded if ambiguous, and SNPs missing from an outcome GWAS are replaced by LD proxies ($r^2 > 0.8$) using the TwoSampleMR package (Hemani et al., 2018). Variants whose strongest association in UK Biobank is with a trait other than depression are removed, since such variants are likely to act on depression through a heritable confounder (Sanderson et al., 2026). This leaves 107 SNPs.

For each SNP we assemble a profile of association estimates across 70 UK Biobank traits with $N > 50{,}000$, after excluding traits with genetic correlation $|r_g| > 0.5$ with depression (so that clusters are not defined by depression itself) and pruning redundant traits (pairwise $|r_g| > 0.5$). Missing cells are imputed by multiple imputation with predictive mean matching ($m = 5$); imputation affects only the clustering step. We cluster the 107 profiles with NAVMIX, a noise-augmented von Mises–Fisher mixture model designed for directional data that allocates variants with incoherent profiles to a separate noise cluster (Grant et al., 2022; Darrous et al., 2024). NAVMIX assigns 39 variants to Cluster 1, 44 to Cluster 2 and 24 to the noise cluster. To characterise each cluster we compute, for every trait t and cluster j, the enrichment ratio $R_{j,t} = \sigma^2_{j,t}/(K^{-1}\sum_k \sigma^2_{k,t})$, where $\sigma^2_{j,t}$ is the mean squared standardised SNP–

trait effect within the cluster. Cluster 1 is enriched for sedentary and neurobehavioural traits (snoring, reaction time, screen time, corneal resistance) and we refer to it as the sedentary cluster; Cluster 2 loads on heterogeneous blood-cell, dietary and medication traits consistent with socioeconomic confounding; the noise cluster is uninterpretable and is used as a placebo instrument.

We then restrict the sedentary cluster to variants that are expression quantitative trait loci (eQTLs) in at least one of thirteen brain regions in GTEx v8 (Consortium, 2020), on the argument that depression liability is most plausibly mediated by neural gene expression. Of the 107 SNPs, 32 are brain eQTLs (6 in the sedentary cluster, 15 in Cluster 2, 11 in the noise cluster). The 6 brain-eQTL sedentary-cluster SNPs — the *brain-expressed sedentary cluster* — form the headline instrument ($F = 44.4$, $R^2 = 0.011\%$); their effect alleles, frequencies, SNP–exposure and SNP–outcome associations and eQTL target genes are reported in Supplementary Section S10. Applying the eQTL filter before clustering yields a coarser structure and attenuated estimates (Supplementary Section S11); we retain the cluster-first ordering because it lets cluster identity be read as a biological hypothesis rather than a residual category.

3.5.2 Estimation and sensitivity analyses

With $\hat{\pi}_l$ the SNP–exposure and $\hat{\Gamma}_l$ the SNP–outcome association for SNP l, both standardised to per-s.d. units using effect-allele frequency and sample size, the inverse-variance weighted (IVW) estimator is $\hat{\beta}_{\mathrm{IVW}} = \sum_l w_l \hat{\Gamma}_l / \hat{\pi}_l \big/ \sum_l w_l$ with $w_l = \hat{\pi}_l^2 / \mathrm{Var}(\hat{\Gamma}_l)$. For each outcome we also report the weighted-median, weighted-mode and MR-PRESSO estimators, which are consistent under different patterns of pleiotropy; Cochran's Q, I^2 and the MR-Egger intercept as heterogeneity and pleiotropy diagnostics; leave-one-SNP-out estimates, scatter and funnel plots; and Steiger tests of the assumed exposure-to-outcome direction (Supplementary Section S9). Instrument validity is benchmarked against the positive and negative controls listed above and against the noise-cluster placebo.

Roughly 60% of UK Biobank participants are common to the exposure and income discovery samples. Sample overlap biases IVW towards the observational association, and selecting instruments in the exposure sample induces winner's curse. We address both with the MR-LAP correction of Mounier and Kutalik (2023), which uses the cross-trait LD-score intercept to correct the IVW estimate, applied to the 6-SNP instrument and to a genome-wide instrument selected by MR-LAP itself, and by re-estimating the income MR with outcome GWAS that exclude UK Biobank (Kweon et al., 2024). With $F = 44.4$ the worst-case overlap bias is a small fraction of the observational estimate (Burgess et al., 2016).

3.5.3 Within-family MR

To rule out effects that run through the family a respondent grew up in (dynastic effects) or through assortative mating, we re-estimate the MR using the family-based GWAS of Guan et al. (2025), which reports both population and direct (within-sibship) SNP–depression effects in UK Biobank. Because direct effects are weakly identified (mean $F \approx 2.2$), we select instruments at $p < 10^{-5}$ in the population arm (80–91 SNPs depending on outcome availability) and use the debiased IVW estimator of Ye et al. (2021), which is consistent

under many weak instruments, with sandwich standard errors. Equality of the population and direct arms is tested by Monte Carlo simulation accounting for their correlation, and sensitivity to residual sample overlap is reported as a function of the exposure–outcome error correlation (Supplementary Section S13).

3.6 PGI as instrument

A polygenic index summarises genetic liability as a weighted sum of allele counts across the genome, $PGI_i = \sum_j \hat{\beta}_j^{\mathrm{GWAS}} x_{ij}$, with weights from an external discovery GWAS. We construct a depression PGI for each genotyped UKHLS participant with PLINK2 following Choi et al. (2020), using genome-wide weights from Baselmans et al. (2019) rather than genome-wide-significant SNPs only, to maximise first-stage strength (Okbay et al., 2016); the discovery sample does not include UKHLS. The PGI trades the interpretability of a small, biologically characterised instrument for power, and is therefore used as a complement to 2SMR.

We estimate by 2SLS (Angrist and Pischke, 2009)

$$y_{i,t} = \beta\, dep_{i,t} + \delta \mathbf{X}_{i,t} + d_t + \epsilon_{i,t}, \qquad dep_{i,t} = \gamma\, PGI_i + \delta' \mathbf{X}_{i,t} + d_t + \nu_{i,t}, \tag{2}$$

on person-waves at ages 25–55 in the same labour-force states as the panel model ($n = 53{,}944$ person-waves from 5,561 individuals; Supplementary Table S3), with wave fixed effects and standard errors clustered by household. To address the independence assumption, $\mathbf{X}_{i,t}$ includes parental education and parental occupation, the first ten genetic principal components, and PGIs for educational attainment and physical health, so that the depression PGI's effect is not carried by family background or by correlated genetic endowments. Sex-specific effects come from a pooled interaction model in which both $dep_{i,t}$ and the instrument are interacted with sex, which pools first-stage power across sexes; average marginal effects and their difference are computed from this model. Mediation is assessed by adding education, first occupation (SOC-90 major group), employment history (lifetime employment rate, long-term sickness, number of job changes), current employment status, current occupation and working hours in sequence, and recording the attenuation of β. Instrument strength (first-stage $F = 1{,}241$), the Anderson–Rubin test, the tF critical values of Lee et al. (2022), a leave-one-household-out jackknife, an OLS comparison, alternative mental-health measures and winsorisation of income are reported in Supplementary Section S14.

3.7 Statistical reporting

Effects are reported per one-s.d. increase in depressive symptoms in s.d. units of the outcome, with 95% confidence intervals. For the four primary outcomes we report unadjusted p-values, since each tests a distinct pre-specified hypothesis; control analyses use a Bonferroni correction across the control set. The four primary outcomes, the instrument-selection rules and the headline specifications were fixed before the 2SMR was run; cluster-specific MR, the brain-eQTL filter, the mediation analysis and the sex-interaction model were developed during the analysis and are labelled as exploratory. Completed STROBE-MR and STROBE checklists are provided in Supplementary Section S18.

3.8 Software

Analyses were run in R 4.5.1. Two-sample MR used TwoSampleMR (Hemani et al., 2018), with GWAS data accessed via ieugwasr; clustering used navmix (Grant et al., 2022); imputation used mice; MR-LAP used MRlap (Mounier and Kutalik, 2023); panel, IV-FE and 2SLS models used fixest, with Sun–Abraham event studies via its `sunab` interface; marginal effects used marginaleffects; PGIs were built with PLINK2. Package versions are listed in Supplementary Section S2.

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Wave 2/3 Nurse assessment interviews were conducted by Natcen. The initial 21 blood analytes were produced by Newcastle upon Tyne Hospitals NHS Foundations Trust and subsequently Royal Free Foundation NHS Trust produced 5 further measures. Genotyping was conducted by the Sanger Institute using the Illumina core and exome chip who also imputed data to the UK10k reference dataset. DNA methylation was assessed using the Illumina Infinium HumanMethylationEPIC BeadChip by the University of Exeter. Proteins were measured by Olink® using the Olink® Target 96 Cardiometabolic and Neurology panels. IP12 dried blood spot analysis was undertaken by MRC Epidemiology Unit Biorepository, Cambridge and hair analysis by Dresden Lab Service, Dresden Germany.

Ethical Approval

Our two-sample MR analysis relied on aggregate GWAS summary statistics data, so no ethics approval or participant consent was required. For the analysis using Understanding Society, the data was obtained under project number Omics #57, and individuals had consented to provide their genetics data for research purposes.

The University of Essex Ethics Committee has approved all data collection on Understanding Society main study, COVID-19 surveys and innovation panel waves, including asking consent for all data linkages except to health records. Approval for the collection of biosocial data by trained nurses in Waves 2 and 3 of the main survey was obtained from the National Research Ethics Service (Understanding Society – UK Household Longitudinal Study: A Biosocial Component, Oxfordshire A REC, Reference: 10/H0604/2).

Pre-registration

This study was not pre-registered. The four primary economic outcomes, the instrument-selection rules, and the main analytic decisions were fixed before running the two-sample MR.

Cluster-specific MR, brain-eQTL filtering, the mediation analysis and the sex-interaction specification were developed iteratively as the analysis progressed, and should be treated as exploratory in the formal STROBE-MR sense.

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Data availability

This study uses publicly available GWAS summary statistics from Baselmans et al. (2019) for the depression exposure, and from Kweon et al. (2024); Ko et al. (2022); Okbay et al. (2022) and the OpenGWAS database for outcome and control phenotypes; full source details and accession identifiers are listed in Supplementary Table S18. Brain eQTL data are from the Genotype-Tissue Expression (GTEx) project (Consortium, 2020). Individual-level UKHLS / Understanding Society data are available from the UK Data Service (Study Number 6614) under standard end-user license. Genotyped UKHLS data are available to bona-fide researchers via application to Understanding Society (Omics project #57). UK Biobank data are available via application to UK Biobank.

Code availability

All analysis code (R, including the NAVMIX clustering pipeline, brain-eQTL filtering, two-sample MR, PGI construction, panel models, and figures) is available at [GitHub URL — to be confirmed], with an archived release on Zenodo (DOI to be issued on acceptance).

Competing interests

The authors declare no competing interests.

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